Exponential and Logarithmic Equations

Section 1: Equations with Like Bases

1 Solve $27^{x-1} = 9^{2x+3}$ for x.

a. -9

b. 9

c. -3

d. 6

2 Solve $5^{x^2-3x} = 625$.

a. $x = -1$ and $x = 4$

b. $x = 1$ and $x = -4$

c. $x = 2$ and $x = -2$

d. $x = 4$ only

3 Solve $\left(\frac{1}{2}\right)^{x+2} = 8^{x-1}$.

a. $1/4$

b. $1/2$

c. $-1/4$

d. 2

Section 2: Equations Requiring Natural Logarithms

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4 Solve $3^{2x} = 7$ for x in exact form.

a. $x = \frac{\ln 7}{2\ln 3}$

b. $x = \frac{\ln 3}{2\ln 7}$

c. $x = \frac{7}{6}$

d. $x = \frac{\ln 7}{\ln 6}$

5 Solve $5e^{3x-2} = 40$ for x.

a. $x = \frac{\ln 8 + 2}{3}$

b. $x = \frac{\ln 8 - 2}{3}$

c. $x = \frac{\ln 40}{15}$

d. $x = 2$

6 Solve $2^{x+1} = 5^{x-1}$ for x.

a. $x = \frac{\ln 2 + \ln 5}{\ln 5 - \ln 2} = \frac{\ln 10}{\ln(5/2)}$

b. $x = \frac{\ln 5}{\ln 2}$

c. $x = 1$

d. $x = \frac{\ln 10}{\ln 10} = 1$

Section 3: Logarithmic Equations and Extraneous Solutions

7 Solve $\log_2(x) + \log_2(x-2) = 3$.

a. $x = 4$ only

b. $x = 4$ and $x = -2$

c. $x = 8$

d. $x = 2$

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8 Solve $\ln(x+5) - \ln(x-1) = \ln(3)$.

a. $x = 4$

b. $x = 3$

c. $x = 2$

d. $x = 5$

9 Why must potential solutions to logarithmic equations always be checked in the original equation?

a. The argument of every logarithm must be strictly positive (> 0)

b. Logarithms cannot have rational answers

c. Extraneous roots always equal 0

d. Base change invalidates solutions

Section 4: Quadratic-Form Exponential Equations

10 Solve $e^{2x} - 5e^{x} + 6 = 0$ for x.

a. $x = \ln 2$ and $x = \ln 3$

b. $x = 2$ and $x = 3$

c. $x = \ln 5$ and $x = \ln 6$

d. $x = e^{2}$ and $x = e^{3}$

11 Solve $e^{2x} - 3e^{x} - 4 = 0$ for x.

a. $x = \ln 4$ only

b. $x = \ln 4$ and $x = \ln(-1)$

c. $x = 4$ and $x = -1$

d. No real solutions

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12 Solve $4^x - 3 \cdot 2^x - 4 = 0$.

a. $x = 2$ only

b. $x = 4$ only

c. $x = 2$ and $x = -1$

d. $x = 1$

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